LOGISTICS RESILIENCE INTELLIGENCE PLATFORM (LRIP)

Version: 1.0

Status: Hackathon MVP + Startup Scale Roadmap

Country: Indonesia

Executive Summary

LRIP is an AI-powered logistics resilience platform designed to predict disruptions, recommend alternative routes, and assist decision-makers in maintaining supply-chain continuity across Indonesia's land and maritime transportation network.

The platform integrates weather, disaster, traffic, vessel, GPS, marketplace pricing, and OSINT data into a unified operational intelligence layer.

The goal is to reduce logistics disruption impacts, improve disaster response, reduce operational costs, and help stabilize commodity prices during crises.

Problem Statement

Indonesia faces persistent logistics challenges:

- Archipelagic geography
- Natural disasters
- Weather disruptions
- Port congestion
- Road bottlenecks
- Fragmented logistics data

Current decision-making is largely reactive.

Organizations must monitor multiple systems manually:

- BMKG
- Maps
- Vessel tracking
- Social media

- News
- GPS systems

No unified predictive intelligence platform currently exists.

Vision

Become Indonesia's national logistics intelligence infrastructure for commercial logistics, humanitarian operations, and disaster-response coordination.

Mission

Predict disruptions before they occur and recommend optimal response actions.

User Segments

Operations Coordinator

Examples:

- Fleet coordinators
- Logistics planners
- NGO coordinators
- Disaster-response coordinators

Goals:

- Reduce delays
- Improve routing
- Increase visibility

Pain Points:

- Uncertain road conditions
- Weather impacts
- Lack of unified monitoring

Decision Maker

Examples:

- Executives
- Government command centers
- Regional directors

Goals:

- Reduce operational costs
- Improve resilience
- Improve disaster preparedness

Pain Points:

- Limited predictive insights
- Poor visibility into systemic risks

Core Value Proposition

Predict logistics disruptions and recommend actions before operational impacts occur.

MVP Scope

Included

- Real-time monitoring dashboard
- AI Copilot
- Route recommendation engine
- Weather integration
- BMKG integration
- AIS vessel tracking
- Driver mobile GPS tracking
- OSINT monitoring
- Risk scoring

Excluded

- Autonomous rerouting
- Direct dispatching
- Satellite integration
- Enterprise ERP integration

Data Sources

Logistics

- AIS vessel APIs
- Driver mobile GPS

Weather

- BMKG
- OpenWeather

Traffic

- Google Maps API

OSINT

- Social media
- News portals
- Community reports

Economic

- Marketplace pricing
- Commodity pricing
- Inflation indicators

Multi-Agent Architecture

Agent 1

Data Collection Agent

Responsibilities:

- API ingestion
- Data normalization
- Data quality validation

Agent 2

OSINT Agent

Responsibilities:

- Flood detection
- Accident detection
- Road closure detection
- Public sentiment monitoring

Agent 3

Prediction Agent

Responsibilities:

- Congestion forecasting
- Weather impact forecasting
- Port delay forecasting

Forecast windows:

- 6 hours
- 12 hours
- 24 hours
- 48 hours

Agent 4

Route Optimization Agent

Responsibilities:

- Alternative route generation
- Risk analysis
- Cost estimation

Outputs:

- ETA
- Fuel impact
- Risk score

Agent 5

Economic Intelligence Agent

Responsibilities:

- Commodity monitoring
- Marketplace monitoring
- Inflation anomaly detection

Agent 6

Decision Support Agent

Responsibilities:

- Generate recommended actions
- Explain reasoning
- Produce executive summaries

AI Architecture

LLM Layer

Primary:

- DeepSeek

Secondary:

- OpenAI/Claude/Gemini

Usage:

- Copilot
- Summaries
- Reasoning

Framework

LangChain

Responsibilities:

- Tool orchestration
- Agent orchestration
- Memory management

RAG

Knowledge Sources:

- SOPs
- Disaster playbooks
- Regulations
- Historical incidents
- Internal documents

Vector Database:

- Qdrant

Memory

Session Memory

Conversation context

Organizational Memory

Company preferences

Examples:

- Preferred routes
- Preferred suppliers
- Historical incidents

Product Features

Dashboard

Displays:

- Route status
- Vessel positions
- Risk maps
- Weather alerts

AI Copilot

Questions:

"Why is delivery delayed?"

"What is the best alternative route?"

"What will happen if Port A closes?"

Risk Engine

Risk categories:

- Weather
- Disaster
- Congestion
- Economic

Output:

Low Medium High Critical

Business Model

Freemium

Purpose:

Marketing and ecosystem growth.

Features:

- Public disruption map
- Weather alerts
- Regional risk index

No:

- AI copilot
- Route optimization
- Prediction engine

Starter

Rp2M-Rp5M/month

Target:

Small logistics operators

Features:

- Dashboard
- Alerts
- Basic monitoring

Professional

Rp15M-Rp30M/month

Features:

- AI Copilot
- Predictions
- Route optimization
- Economic intelligence

Enterprise

Rp100M+/year

Features:

- API access
- SSO
- Custom integrations
- SLA

Government

Custom procurement contracts

Revenue sources:

- Provincial governments
- Ministries
- BNPB
- BPBD

Revenue Streams

1. SaaS subscriptions

2. Enterprise contracts

3. Government contracts

4. API licensing

5. Analytics reports

6. Future data marketplace commissions

Cost Structure

MVP

Cloud:

Rp3M-Rp10M/month

AI:

Rp2M-Rp20M/month

Database:

Rp1M-Rp3M/month

Monitoring:

Rp500K-Rp2M/month

Total:

Rp10M-Rp40M/month

Scaling Model

Variable Costs:

- API calls
- GPS pings
- AI requests

Expected scaling:

Near-linear during early growth.

Margins improve after:

- caching
- batch processing
- model routing

Success Metrics

Operational

Route optimization acceptance rate

Target: 70%

Prediction accuracy

Target: 80%

Alert precision

Target: 85%

Business

Monthly recurring revenue

Customer retention

Government pilot contracts

Enterprise conversion rate

Long-Term Roadmap

Phase 1

MVP

0-6 months

Phase 2

Enterprise integrations

6-18 months

Phase 3

National logistics intelligence platform

18-36 months

Phase 4

ASEAN expansion

36+ months